

The graphwise lattice-to-continuum lemma for the Brazovskii correlators

TECT verification-first repository · claim A3-PERTURBATIVE-CONTINUUM-CORRELATORS

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1. Purpose and scope

This note closes the gap **A3-GRAPHWISE-CONVERGENCE** left open in v1.0: it proves that every connected perturbative amplitude of the scalar Brazovskii theory converges, lattice-to-continuum, to its continuum value, and that the lattice spacing $a \rightarrow 0$ and the cutoff $\Lambda \rightarrow \infty$ are a single limit. Together with A3-UV (the uniform bound and Weinberg absolute convergence) this upgrades A3-PERTURBATIVE-CONTINUUM-CORRELATORS to T6 in the scalar/ $d=3/\mu^2 > 0$ perturbative scope. The argument is standard lattice regularisation + dominated convergence (Reed–Simon I; Glimm–Jaffe). Numerical ingredients: [codes/foundations/a3_graphwise_convergence_checks.py](#) (7/7 PASS).

2. The lattice regulator

For spacing a , work on the Brillouin zone $\text{BZ}_a = [-\pi/a, \pi/a]^3$ with the lattice momentum and kernel

$$\hat{q}_j(q) = \frac{2}{a} \sin \frac{aq_j}{2}, \quad K_a(q) = \mu^2 + Y(\hat{q}(q)^2 - q_0^2)^2, \quad G_a = \frac{1}{K_a}, \quad (1)$$

so the cutoff is $\Lambda_a = \pi/a$. A connected graph $\mathcal{G}$ (V vertices, I lines, $L = I - V + 1$ loops, constant ϕ^4/ϕ^6 couplings) has amplitude

$$\mathcal{A}_{\mathcal{G},a}(p) = \int_{(\mathbb{R}^3)^L} \prod_{\ell=1}^L \frac{d^3 k_\ell}{(2\pi)^3} \mathbf{1}_{\text{BZ}_a}(k) \prod_{i=1}^I G_a(q_i(k, p)), \quad (2)$$

the q_i linear in (k, p) ; $\mathcal{A}_{\mathcal{G}}$ is the same with $\mathbf{1}_{\text{BZ}_a} \rightarrow 1$ and $G_a \rightarrow G = 1/K$.

3. Pointwise convergence

For each fixed q , $\hat{q}_j(q) = \frac{2}{a} \sin \frac{aq_j}{2} \rightarrow q_j$ as $a \rightarrow 0$ (assert `lattice_momentum_pointwise_convergence`), hence $K_a(q) \rightarrow K(q)$ and $G_a(q) \rightarrow G(q)$ pointwise; and $\mathbf{1}_{\text{BZ}_a}(k) \rightarrow 1$ for every fixed k since $\Lambda_a \rightarrow \infty$. Thus the integrand converges pointwise a.e. to $\prod_i G(q_i)$.

4. Uniform q^{-4} bound

On BZ_a one has $aq_j/2 \in [-\pi/2, \pi/2]$, where $\sin x/x \geq 2/\pi$, so $|\hat{q}_j| \geq \frac{2}{\pi}|q_j|$ and $\hat{q}^2 \geq \frac{4}{\pi^2}q^2$ (assert `lattice_momentum_lower_bound`). Hence for $|q|$ beyond the shell $K_a(q) \geq Y(\hat{q}^2 - q_0^2)^2 \geq c q^4$, and near the shell $K_a \geq \mu^2 > 0$; combining,

$$G_a(q) = \frac{1}{K_a(q)} \leq \frac{C}{(1 + |q|)^4} \quad \text{uniformly in } a, \quad (3)$$

with $C = C(\mu^2, Y, q_0)$ independent of a (assert `uniform_q4_dominating_bound`: $\sup_q G_a(1 + |q|)^4$ varies by $< 3\%$ over $a \in [0.03, 1]$). The scalar kernel has a unique minimum on the shell $\hat{q}^2 = q_0^2$, not at the BZ corners (K_a there $\sim 2.6 \times 10^6 \mu^2$): no spurious doubler (assert `no_scalar_doubler`).

5. Dominated convergence

By (3) the integrand is dominated by $\Phi(k, p) := \prod_{i=1}^I C(1 + |q_i(k, p)|)^{-4}$, independent of a . By A3-UV every subgraph has Weinberg degree $D' = 3 - 3V' - I' < 0$, so $\Phi \in L^1((\mathbb{R}^3)^L)$ for fixed compact p (Weinberg's theorem; assert `dominating_function_integrable_weinberg`). Dominated convergence then exchanges the limit and the loop integrations:

$$\lim_{a \rightarrow 0} \mathcal{A}_{\mathcal{G},a}(p) = \mathcal{A}_{\mathcal{G}}(p), \quad (4)$$

uniformly on compact external momenta (the bound Φ is p -locally uniform).

6. Regulator matching and the single limit

Because $\Lambda_a = \pi/a$, the limits $a \rightarrow 0$ and $\Lambda \rightarrow \infty$ coincide: there is one limit, taken along the lattice family (assert `regulator_matching_single_limit`). No separate matching of a momentum cutoff to the continuum kernel is needed.

Theorem (A3-GRAPHWISE-CONVERGENCE). *In the scalar Brazovskii theory ($d = 3$, $\mu^2 > 0$, quartic kernel) with the lattice regulator $\hat{q}_j = \frac{2}{a} \sin \frac{aq_j}{2}$, every connected perturbative amplitude satisfies $\lim_{a \rightarrow 0} \mathcal{A}_{\mathcal{G},a}(p) = \mathcal{A}_{\mathcal{G}}(p)$, uniformly on compact external momenta, with $a \rightarrow 0 \equiv \Lambda \rightarrow \infty$.*

Corollary (A3-PERTURBATIVE-CONTINUUM-CORRELATORS, T6). Each order of the perturbative correlators $\langle \mathcal{O} \rangle_{\Lambda,a}$ (a finite sum of amplitudes $\mathcal{A}_{\mathcal{G},a}$) converges to its continuum value as $a \rightarrow 0$; the perturbative correlators have a cutoff-independent continuum limit.

7. Devil's-advocate (three objections)

- (α) “Pointwise convergence of the integrand is not enough.” DISMISSED: the proof is dominated convergence, not mere pointwise — the uniform a -independent dominating function Φ (3) and its integrability (Weinberg, §5) are exactly the DCT hypotheses.
- (β) “Lattice artifacts / doublers near the BZ edge.” DISMISSED: the scalar kernel is minimised only on the shell (no doubler), and the edge region is controlled by the same uniform $(1 + |q|)^{-4}$ bound, whose integral is finite; the indicator $\mathbf{1}_{\text{BZ}_a}$ converges to 1 a.e.
- (γ) “This is order-by-order, not the full series.” VALID with mitigation: the claim is the *perturbative* continuum limit (each order), which is what is stated; resummation / the constructive non-perturbative limit remains the separate target $\text{A3}_{\text{constructive}}$.

Quantitative sanity check (mandatory): $|\hat{q}| \geq (2/\pi)|q|$ verified on BZ_a ; $\hat{q} \rightarrow q$ with $\max |\hat{q} - q| \rightarrow 5 \times 10^{-4}$; the uniform constant $\sup_q G_a(1 + |q|)^4 \in [1600, 1654]$ (a -independent to 3%); $\Lambda_a = \pi/a \rightarrow \infty$; BZ-corner $K_a/\mu^2 \sim 5 \times 10^8$ (no doubler); integrated tadpole instance converges to 4.96×10^{-4} (7/7 checks).

8. Result footer

Result ID: A3-PERTURBATIVE-CONTINUUM-CORRELATORS (T6)
Precise statement: With the lattice regulator $\text{qhat}=(2/a)\sin(aq/2)$, every connected perturbative amplitude converges $\lim_{\{a \rightarrow 0\}} A_{\{G,a\}} = A_G$ (uniformly on compact p), $a \rightarrow 0 \equiv \Lambda \rightarrow \infty$; hence the perturbative

correlators have a cutoff-independent continuum
 limit. Gate A3-GRAPHWISE-CONVERGENCE CLOSED.
 Scope: scalar, $d=3$, $\mu^2>0$, PERTURBATIVE (order by order).
 Dependencies: A1-KERNEL-CONV; A3-UV-SUPERRENORMALISABILITY
 (uniform q^{-4} bound + Weinberg integrability).
 Method: lattice regulator + dominated convergence.
 Evidence grade: PROVED CONDITIONAL (analytic) + EXECUTED (7/7).
 Reproduction command: python codes/foundations/
 a3_graphwise_convergence_checks.py
 Expected output: 7/7 PASS; exit 0.
 Falsification gate: a connected graph G with $A_{\{G,a\}}$ not $\rightarrow$ A_G under
 this regulator; or failure of the uniform
 $(1+|q|)^{-4}$ bound on BZ_a .
 Tier before / after: T3 PROOF SKETCH / T6 PROVED CONDITIONAL.
 No-overclaim statement: Order-by-order only; NOT resummation; NOT the
 constructive/non-perturbative limit; not Class-II;
 not $\mu^2<0$.
 Next required action: Operator sign-off; then A3_constructive and the
 physical production-Hessian mainline.